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Prevalence of pretreatment drug resistance in persons initiating and reinitiating antiretroviral therapy in Sri Lanka: results from a national representative survey

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Background: We conducted a nationwide cross-sectional study to estimate pretreatment drug resistance (PDR) prevalence in adults initiating ART in Sri Lanka following the WHO's recommendations.

Methods: HIV drug resistance was determined on dried blood spots (DBSs) using population-based sequencing of the protease and reverse transcriptase genes and interpretation was based on Stanford HIVdb v9.0. Analyses were weighted to adjust for multistage sampling and genotypic failure rate. We used logistic regression to assess differences between groups.

Results: Overall, in 10% (15 of 150) of patients initiating ART, HIV drug resistance mutations were detected. The prevalence of resistance to NNRTI drugs efavirenz/nevirapine was 8.4% (95% CI 4.6–15.0) but differed among those reporting having prior antiretroviral (ARV) exposure (24.4%, 95% CI 13.8–39.5) compared with 4.6% (95% CI 1.6–12.8) for those reporting as being ARV naive (OR 4.6, 95% CI 1.3–16.6, $P=0.021$). PDR to efavirenz/nevirapine was also nearly twice as high among women (14.1%, 95% CI 6.1–29.4) compared with men (7.0%, 95% CI 3.1–14.7) ($P=0.340$) and three times high among heterosexuals (10.4%, 95% CI 2.4–35.4) compared with MSM (3.8%, 95% CI 1.1–12.7) ($P=0.028$). NRTI PDR prevalence was 3.8% (95% CI 1.1–12.1) and no PI PDR was observed in the study.

Conclusions: A high prevalence of efavirenz/nevirapine PDR was reported, especially in patients with prior ARV exposure, in women and those reporting being heterosexual. These findings highlight the need to fast-track the transition to the WHO-recommended dolutegravir-based first-line ART.

Introduction

The global goal for achieving HIV epidemic control by the year 2030 is predicated upon widescale access and use of efficacious ART. The UNAIDS projects that having 95% of HIV patients knowing their status and 95% of those who know their status on treatment and 95% of those on treatment achieving viral load suppression by 2025, will lead to eliminating HIV as a public health threat by 2030.¹ Significant strides have been made towards the attainment of this goal and by the end of 2021, 69% of all people living with HIV were on treatment and virally suppressed but this fell short of the 86% UNAIDS 2025 target.²

The emergence of HIV drug resistance is, however, a threat to the achievements of the global goals. In the 2021 WHO HIV drug resistance report, 16 of 20 countries reported having

pretreatment drug resistance (PDR) levels of >10% to the previously commonly used NNRTIs, efavirenz and/or nevirapine, among adults newly initiating or reinitiating first-line ART.³ PDR to NNRTIs has been associated with poor virological outcomes, impaired immune recovery, and reduced durability of NNRTI-based regimens.^{4,5} A modelling study predicted an increase in HIV incidence, mortality and overall programmatic costs attributed to high levels of PDR.⁵ To prevent the impact of PDR, the WHO recommended the use of dolutegravir-based ART as the preferred first-line ART regimen in 2018.⁶ A generic fixed dose combination of tenofovir disoproxil fumarate, lamivudine and dolutegravir 300/300/50 mg ('TLD') has been made available at an affordable price through the Medicines Patent Pool for low- and middle-income countries. However, access may still be a barrier due to differential pricing, especially among

middle-income and high-income countries.⁷ Sri Lanka has made significant progress towards epidemic control but there are still significant gaps in the first and third UNAIDS targets with the overall 95-95-95 progress being at 51-95-84 based on 2019 data.⁸ At the time of the survey, the majority of those on treatment were on an NNRTI-containing regimen. We thus evaluated the prevalence of PDR in Sri Lanka in order to facilitate the transition to dolutegravir-based therapy with the aim of improving treatment outcomes.

Methods

A cross-sectional survey was carried out following the WHO-recommended methods for PDR surveys using a census of all clinics⁹ after excluding ART facilities with a small cohort of people living with HIV newly initiating ART. A sample size of 163 participants was estimated based on WHO-recommended assumptions. The sample was then assigned proportionally to the cohort of people living with HIV newly initiating ART in each of the 18 selected ART facilities. Participants were recruited at the facilities if they were initiating ART for the first time or re-initiating ART if they had previously stopped for more than 3 months. Eligible individuals were consecutively enrolled from January 2021 to July 2021. Due to challenges in collection of samples and shipment logistics impacted by the COVID pandemic, attempts were made to enrol all eligible cases in the facilities.

After obtaining consent from participants for blood collection, 7 mL of venous EDTA whole blood was collected from all study participants and shipped to the national reference laboratory (NRL) at Colombo within 24 h under cold storage where dried blood spots (DBSs) were prepared in accordance with WHO guidelines. DBSs were then shipped to the WHO-designated laboratory at National AIDS Research Institute (NARI) in India, where genotyping of the protease and reverse transcriptase *pol* genes was done using a validated in-house Sanger-based assay. HIV drug resistance was predicted using the Stanford HIVdb algorithm, v9.0. Sequences classified as having low-, intermediate- or high-level resistance were classified as resistant. HIVDR prevalence estimates were assessed using the 'svy' utilities in Stata (Stata 12; StataCorp, College Station, TX, USA). Analyses were weighted to adjust for multistage sampling and genotypic failure rate. We used logistic regression to assess differences between groups.

Ethics

The study was approved by the national ethical research committee (ERC/PGIM/2020/182). Participants aged >18 years provided written informed consent at study enrolment. All parents and legal guardians gave written permission for participation of minors (aged 15–18 years), who also gave assent, as per local guidelines.

Results

In total, 194 ART initiators were enrolled in the study, the majority of whom were male (78.6%, 95% CI 69.4–87.9) and >25 years old (80.0%, 95% CI 74.2–85.3) (Table 1). ART was initiated by 25.6% (95% CI 17.1–34.1) with a CD4 count of <200 cells/mm³, while 18.2% (95% CI 9.9–26.6) reported having prior exposure to antiretrovirals (ARVs). The proportion of those reporting prior ARV exposure was higher in women versus men (25.0% versus 19.1%, $P=0.427$) and heterosexual versus MSM (27.0% versus 17.6%, $P=0.129$) but the differences were not significant.

Twelve samples were rejected at NARI as they had been exposed to room temperature for >14 days due to shipment

Table 1. Demographic and clinical characteristics of ART initiators, Sri Lanka, 2021

	ART initiators (N=194)	
	n	% (95% CI) ^a
Gender		
Female	36	21.0 (11.7–30.2)
Male	157	78.6 (69.4–87.9)
Others	1	<0.5
Age (years)		
≤25	36	41.9 (35.5–48.3)
>25	157	20.2 (14.7–25.8)
CD4 count (cells/mm ³)		
<200	53	80.0 (74.2–85.3)
≥200	117	385.8 (324.9–446.7)
Perceived HIV risk factor		
Heterosexual	63	29.1 (20.0–38.3)
Bisexual	19	70.9 (61.7–80.0)
FSW	1	
Client of FSW	2	34.1 (23.6–44.5)
MSM	67	8.1 (2.6–13.7)
IVDU	1	0.1 (0–0.2)
MTCT	1	0.04 (0–1.7)
HIV-positive partner	2	34.2 (24.1–44.2)
Various combinations	6	0.04 (0–0.1)
Unknown	28	1.2 (0–3.7)
Others	4	1.2 (0–3.2)
Prior ARV exposure		
Yes	40	2.3 (0.8–3.8)
No	154	13.9 (3.1–24.7)
Type of prior ARV exposure		
Previous ART use	37	2.4 (0–5.9)
Unknown	3	18.2 (9.9–26.6)
NNRTI only	23	81.2 (73.4–90.1)
PI	10	92.5 (84.8–100)
INI	3	7.5 (0–15.2)
Unknown	1	65.4 (46.7–84.2)
Planned regimen		
EFV-based ART	122	24.0 (7.3–40.6)
LPV-r/ATV-r	8	8.2 (0–16.8)
DTG-based ART	51	2.4 (0–8.7)
RAL-based ART	8	69.9 (50.9–89.0)
Unknown	5	4.1 (1.0–7.7)
		21.3 (7.9–34.6)
		2.9 (0–6.8)
		1.8 (0–4.6)

ATV-r, atazanavir-boosted ritonavir; EFV, efavirenz; DTG, dolutegravir; FSW, female sex worker; INI, integrase inhibitors; LPV-r, lopinavir-boosted ritonavir; MTCT, mother-to-child transmission; NVP, nevirapine; RAL, raltegravir.

^aStudy design-weighted proportions and 95% CIs. ART initiators were defined as people living with HIV initiating or reinitiating a first-line regimen.

logistical challenges relating to COVID-19 restrictions. Of the 181 samples accepted for genotyping, 150 (83%) had a successful genotype. The overall prevalence of resistance to efavirenz/nevirapine was 8.4% (95% CI 4.6–15.0) but differed among those reporting having prior ARV exposure (24.4%, 95% CI 13.8–39.5) compared with those reporting to be ARV naive (4.6%, 95% CI 1.6–12.8) (OR 4.6, 95% CI 1.3–16.6, $P=0.021$). PDR to efavirenz/

Table 2. Prevalence of pretreatment HIV drug resistance among ART initiators, Sri Lanka, 2021

Drug class	Drug	All		Women		Men		ARV naive		With prior ARV exposure	
		n/N	% (95% CI) ^a	n/N	% (95% CI) ^a	n/N	% (95% CI) ^a	n/N	% (95% CI) ^a	n/N	% (95% CI) ^a
Any	Any HIVDR	15/150	11.2 (6.6–18.5)	5/28	14.1 (6.1–29.4)	10/121	10.5 (5.4–19.6)	7/116	8.0 (3.4–17.8)	8/34	24.4 (13.8–39.5)
NRTI	Any	4/150	3.8 (1.1–12.1)	0/28	—	4/121	4.8 (1.4–15.2)	4/116	4.5 (1.5–13.2)	0/34	—
	ABC	0/150	—	0/28	—	0/121	—	0/116	—	0/34	—
	3TC or FTC	0/150	—	0/28	—	0/121	—	0/116	—	0/34	—
	TDF	0/150	—	0/28	—	0/121	—	0/116	—	0/34	—
	ZDV	2/150	1.6 (0.3–8.0)	0/28	—	2/121	2.0 (0.4–10.3)	2/116	2.0 (0.4–9.6)	0/34	—
NNRTI	EFV or NVP	12/150	8.4 (4.6–15.0)	5/28	14.1 (6.1–29.4)	7/121	7.0 (3.1–14.7)	4/116	4.6 (1.6–12.8)	8/34	24.4 (13.8–39.5)
	DOR	5/150	2.5 (0.8–8.0)	1/28	2.3 (0.4–12.7)	4/121	2.6 (0.9–7.6)	1/116	0.6 (0.1–3.2)	4/34	10.4 (4.2–23.4)
	ETR	2/150	1.0 (0.2–4.7)	1/28	2.3 (0.4–12.7)	1/121	0.6 (0.1–2.9)	1/106	0.6 (0.1–3.2)	1/34	2.5 (0.7–8.4)
	RPV	4/150	2.0 (0.7–6.1)	2/28	4.6 (0.7–23.7)	2/121	1.4 (0.4–4.5)	1/106	0.6 (0.1–3.2)	3/34	7.9 (3.1–18.6)
PI/r	ATV/r, DRV/r or LPV/r	0/150	—	0/28	—	0/121	—	0/106	—	0/34	—
	ATV/r	0/150	—	0/28	—	0/121	—	0/106	—	0/34	—
	DRV/r	0/150	—	0/28	—	0/121	—	0/106	—	0/34	—
	LPV/r	0/150	—	0/28	—	0/121	—	0/106	—	0/34	—

HIVDR was defined as the presence of a penalty score ≥ 15 using the Stanford HIVdb algorithm.

ABC, abacavir; ATV/r, atazanavir/ritonavir; BIC, bictegravir; CAB, cabotegravir; DOR, doravirine; DRV/r, darunavir/ritonavir; DTG, dolutegravir; EFV, efavirenz; ETV, etravirine; EVG, elvitegravir; INSTI, integrase strand transfer inhibitor; LPV/r, lopinavir/ritonavir; —, no data; NVP, nevirapine; PI/r, boosted PI; RAL, raltegravir; RPV, rilpivirine; TDF, tenofovir; ZDV, zidovudine; 3TC/FTC, lamivudine/emtricitabine.

^aStudy design-weighted proportion and 95% CI.

nevirapine was also nearly twice as high among women (14.1%, 95% CI 6.1–29.4) compared with men (7.0%, 95% CI 3.1–14.7) but the difference was not significant ($P=0.340$). The prevalence of PDR to efavirenz/nevirapine was also three times higher among heterosexuals compared with MSM (10.4%, 95% CI 2.4–35.4) and compared with men (3.8%, 95% CI 1.1–12.7) ($P=0.028$).

Overall PDR prevalence to any drug was 11.2% (95% CI 6.6–18.5) while that to NRTIs was low at 3.8% (95% CI 1.1–12.1) and no PI PDR was observed in the study (Table 2). PDR to NRTIs was driven mainly by zidovudine resistance due to the presence of thymidine analogue mutations (TAMs). Prevalence of PDR to second-generation NNRTIs was also low, ranging from 2.5% (95% CI 0.8–8.0) for doravirine, 2.0% (95% CI 0.7–6.1) for rilpivirine and 1.0% (95% CI 0.2–4.7) for etravirine. K103N/H/T was the most commonly observed NNRTI resistance mutation. Of the 12 patients with PDR to efavirenz/nevirapine, 8 had been planned to be initiated on efavirenz/nevirapine, 3 on dolutegravir and 1 on a ritonavir-boosted lopinavir-containing regimen (Table 1).

Discussion

The prevalence of pretreatment HIV drug resistance to efavirenz or nevirapine was near borderline (8.4%) at the WHO-recommended cut-off of 10% that necessitates moving away from an efavirenz-based regimen. However, the prevalence of PDR to efavirenz/nevirapine had exceeded 10% in some subpopulations. Among those initiating treatment with prior ARV exposure, PDR to efavirenz/nevirapine was much higher at 24%.

This group represents about 18% of all patients initiating treatment in Sri Lanka. PDR to efavirenz/nevirapine was also above 10% among women (14.1%). The proportion of women was, however, underrepresented in the survey because of the low HIV prevalence in this group in Sri Lanka. Overall, these findings highlight the need to accelerate the adoption of WHO's and Sri Lanka's 2020 recommendations to scale-up dolutegravir-based therapy so as to minimize the impact of PDR on patient outcomes and ensure the country's attainment of the global targets on viral suppression.^{10,11} This is especially critical noting that 67% of those with PDR to efavirenz/nevirapine were planned to be initiated on a regimen that they were already resistant to, thus highlighting fast-tracking transition to dolutegravir-based ART.

About one in five patients initiating treatment had been previously exposed to ARVs for treatment. This high number of patients reinitiating treatment after previous disengagement from care highlights the need to strengthen systems to minimize attrition from care by implementing vigorous local retention strategies as also recommended by WHO.¹¹ Moreover, since patients re-engaging in care are likely to have poor treatment outcomes,^{4,12} there is need for close monitoring of this group including addressing the factors associated with the initial default from care.¹³ Contrary to previous reports was the nearly twice as high level of PDR among heterosexuals compared with MSM.¹⁴ This is likely due to a concentration of interventions among MSM due to the high level of HIV prevalence in this group. These findings, however, highlight the need to not only focus on key populations but also on the general population.

We observed a low prevalence of NRTI resistance as well as resistance to the second-generation NNRTIs. These findings give reassurance for the potential efficacy of Truvada (tenofovir and emtricitabine)-based pre-exposure prophylaxis as it becomes rolled out among key populations in Sri Lanka.¹⁵

Besides PDR, we observed that about one in every three patients starting treatment had advanced disease with CD4 counts of <200 cells/mm³. This observation is concerning, as patients with advanced disease have an increased risk of developing AIDS and death. Moreover, patients initiating treatment with low CD4 counts are less likely to achieve optimal immune recovery, even while on treatment predisposing them to the risk of long-term AIDS and non-AIDS complications, and death.^{16,17} This highlights an urgent need to strengthen and innovate strategies for HIV diagnosis and linkage to care and treatment, including self-testing and assisted partner notification, to increase the rates of early diagnosis of HIV infection and rapid treatment initiation. Moreover, there is the need to accelerate the implementation of WHO's and Sri Lanka's advanced disease guidelines, a package of screening, prophylaxis, rapid ART initiation and intensified adherence interventions, to help reduce morbidity and mortality.¹⁸

Overall, our observation of a high level of PDR to efavirenz/nevirapine in subpopulations such as women and those initiating treatment with prior ARV exposure calls for accelerated transition to TLD in line with the current Sri Lanka guideline. If rapid transition is not feasible, the use of PDR genotyping testing, coupled with screening of patients with prior ARV exposure, would be a suitable alternative in the interim.

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Transparency declarations

All authors declare that they have no conflict of interest.

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